



changes to, or deletion of data while retaining previous and original data. It should be possible to associate all data and changes to data with the persons making those changes, and changes should be dated and time stamped (time and time zone where applicable). The reason for any change, should also be recorded. The items included in the audit trail should be those of relevance to permit reconstruction of the process or activity.

Audit trails (identified by risk assessment as required) should be switched on. Users should not be able to amend or switch off the audit trail. Where a system administrator amends, or switches off the audit trail a record of that action should be retained.

The relevance of data retained in audit trails should be considered by the organisation to permit robust data review/verification. It is not necessary for audit trail review to include every system activity (e.g. user log on/off, keystrokes etc.).

Where relevant audit trail functionality does not exist (e.g. within legacy systems) an alternative control may be achieved for example defining the process in an SOP, and use of log books. Alternative controls should be proven to be effective.

Where add-on software or a compliant system does not currently exist, continued use of the legacy system may be justified by documented evidence that a compliant solution is being sought and that mitigation measures temporarily support the continued use.¹

Routine data review should include a documented audit trail review where this is determined by a risk assessment. When designing a system for review of audit trails, this may be limited to those with GXP relevance. Audit trails may be reviewed as a list of relevant data, or by an 'exception reporting' process. An exception report is a validated search tool that identifies and documents predetermined 'abnormal' data or actions, that require further attention or investigation by the data reviewer.

Reviewers should have sufficient knowledge and system access to review relevant audit trails, raw data and metadata (see also 'data governance').

Where systems do not meet the audit trail and individual user account expectations, demonstrated progress should be available to address these shortcomings. This should either be through add-on software that provides these additional functions or by an upgrade to a compliant system. Where remediation has not been identified or subsequently implemented in a timely manner a deficiency may be cited.

6.14. Electronic signatures

A signature in digital form (bio-metric or non-biometric) that represents the signatory. This should be equivalent in legal terms to the handwritten signature of the signatory.

The use of electronic signatures should be appropriately controlled with consideration given to:

- How the signature is attributable to an individual.

¹ It is expected that GMP facilities with industrial automation and control equipment/ systems such as programmable logic controllers should be able to demonstrate working towards system upgrades with individual login and audit trails (reference: Art 23 of Directive 2001/83/EC).



- How the act of 'signing' is recorded within the system so that it cannot be altered or manipulated without invalidating the signature or status of the entry.
- How the record of the signature will be associated with the entry made and how this can be verified.
- The security of the electronic signature i.e. so that it can only be applied by the 'owner' of that signature.

It is expected that appropriate validation of the signature process associated with a system is undertaken to demonstrate suitability and that control over signed records is maintained. Where a paper or pdf copy of an electronically signed document is produced, the metadata associated with an electronic signature should be maintained with the associated document.

The use of electronic signatures should be compliant with the requirements of international standards. The use of advanced electronic signatures should be considered where this method of authentication is required by the risk assessment. Electronic signature or E-signature systems must provide for "signature manifestations" i.e. a display within the viewable record that defines who signed it, their title, and the date (and time, if significant) and the meaning of the signature (e.g. verified or approved).

An inserted image of a signature or a footnote indicating that the document has been electronically signed (where this has been entered by a means other than the validated electronic signature process) is not adequate. Where a document is electronically signed then the metadata associated with the signature should be retained.

For printed copies of electronically signed documents refer to True Copy section.

Expectations for electronic signatures associated with informed consent (GCP) are covered in alternative guidance (MHRA/HRA DRAFT Guidance on the use of electronic consent).

6.15. Data review and approval

The approach to reviewing specific record content, such as critical data and metadata, cross-outs (paper records) and audit trails (electronic records) should meet all applicable regulatory requirements and be risk-based.

There should be a procedure that describes the process for review and approval of data. Data review should also include a risk-based review of relevant metadata, including relevant audit trails records. Data review should be documented and the record should include a positive statement regarding whether issues were found or not, the date that review was performed and the signature of the reviewer.

A procedure should describe the actions to be taken if data review identifies an error or omission. This procedure should enable data corrections or clarifications to provide visibility of the original record, and traceability of the correction, using ALCOA principles (see 'data' definition).

Where data review is not conducted by the organisation that generated the data, the responsibilities for data review must be documented and agreed by both parties. Summary



reports of data are often supplied between organisations (contract givers and acceptors). It must be acknowledged that summary reports are limited and critical supporting data and metadata may not be included.

Many software packages allow configuration of customised reports. Key actions may be incorporated into such reports provided they are validated and locked to prevent changes. Automated reporting tools and reports may reduce the checks required to assure the integrity of the data.

Where summary reports are supplied by a different organisation, the organisation receiving and using the data should evaluate the data provider's data integrity controls and processes prior to using the information.

- Routine data review should consider the integrity of an individual data set e.g. is this the only data generated as part of this activity? Has the data been generated and maintained correctly? Are there indicators of unauthorised changes?
- Periodic audit of the data generated (encompassing both a review of electronically generated data and the broader organisational review) might verify the effectiveness of existing control measures and consider the possibility of unauthorised activity at all interfaces, e.g. have there been IT requests to amend any data post review? Have there been any system maintenance activities and has the impact of that activity been assessed?

6.16. Computerised system user access/system administrator roles

Full use should be made of access controls to ensure that people have access only to functionality that is appropriate for their job role, and that actions are attributable to a specific individual. Companies must be able to demonstrate the access levels granted to individual staff members and ensure that historical information regarding user access level is available. Where the system does not capture this data, then a record must be maintained outside of the system. Access controls should be applied to both the operating system and application levels. Individual login at operating system level may not be required if appropriate controls are in place to ensure data integrity (e.g. no modification, deletion or creation of data outside the application is possible).

For systems generating, amending or storing GXP data shared logins or generic user access should not be used. Where the computerised system design supports individual user access, this function must be used. This may require the purchase of additional licences. Systems (such as MRP systems) that are not used in their entirety for GXP purposes but do have elements within them, such as approved suppliers, stock status, location and transaction histories that are GXP applicable require appropriate assessment and control.

It is acknowledged that some computerised systems support only a single user login or limited numbers of user logins. Where no suitable alternative computerised system is available, equivalent control may be provided by third-party software or a paper-based method of providing traceability (with version control). The suitability of alternative systems should be justified and documented. Increased data review is likely to be required for hybrid systems



because they are vulnerable to non-attributable data changes. It is expected that companies should be implementing systems that comply with current regulatory expectations².

System administrator access should be restricted to the minimum number of people possible taking account of the size and nature of the organisation. The generic system administrator account should not be available for routine use. Personnel with system administrator access should log in with unique credentials that allow actions in the audit trail(s) to be attributed to a specific individual. The intent of this is to prevent giving access to users with potentially a conflict of interest so that they can make unauthorised changes that would not be traceable to that person.

System Administrator rights (permitting activities such as data deletion, database amendment or system configuration changes) should not be assigned to individuals with a direct interest in the data (data generation, data review or approval).

Individuals may require changes in their access rights depending on the status of clinical trial data. For example, once data management processes are complete, the data is 'locked' by removing editing access rights. This should be able to be demonstrated within the system.

6.17. Data retention

Data retention may be for archiving (protected data for long-term storage) or backup (data for the purposes of disaster recovery).

Data and document retention arrangements should ensure the protection of records from deliberate or inadvertent alteration or loss. Secure controls must be in place to ensure the data integrity of the record throughout the retention period and should be validated where appropriate (see also data transfer/migration).

Data (or a true copy) generated in paper format may be retained by using a validated scanning process provided there is a documented process in place to ensure that the outcome is a true copy.

Procedures for destruction of data should consider data criticality and where applicable legislative retention requirements.

² It is expected that GMP facilities with industrial automation and control equipment/ systems such as programmable logic controllers should be able to demonstrate working towards system upgrades with individual login and audit trails (reference: Art 23 of Directive 2001/83/EC).



6.17.1. Archive

A designated secure area or facility (e.g. cabinet, room, building or computerised system) for the long term, retention of data and metadata for the purposes of verification of the process or activity.

Archived records may be the original record or a 'true copy' and should be protected so they cannot be altered or deleted without detection and protected against any accidental damage such as fire or pest.

Archive arrangements must be designed to permit recovery and readability of the data and metadata throughout the required retention period. In the case of archiving of electronic data, this process should be validated, and in the case of legacy systems the ability to review data periodically verified (i.e. to confirm the continued support of legacy computerised systems). Where hybrid records are stored, references between physical and electronic records must be maintained such that full verification of events is possible throughout the retention period.

When legacy systems can no longer be supported, consideration should be given to maintaining the software for data accessibility purposes (for as long possible depending upon the specific retention requirements). This may be achieved by maintaining software in a virtual environment.

Migration to an alternative file format that retains as much as possible of the 'true copy' attributes of the data may be necessary with increasing age of the legacy data. Where migration with full original data functionality is not technically possible, options should be assessed based on risk and the importance of the data over time. The migration file format should be selected considering the balance of risk between long-term accessibility versus the possibility of reduced dynamic data functionality (e.g. data interrogation, trending, re-processing etc). It is recognised that the need to maintain accessibility may require migration to a file format that loses some attributes and/or dynamic data functionality (see also 'Data Migration').

6.17.2. Backup

A copy of current (editable) data, metadata and system configuration settings maintained for recovery including disaster recovery.

Backup and recovery processes should be validated and periodically tested. Each back up should be verified to ensure that it has functioned correctly e.g. by confirming that the data size transferred matches that of the original record.

The backup strategies for the data owners should be documented.

Backups for recovery purposes do not replace the need for the long term, retention of data and metadata in its final form for the purposes of verification of the process or activity.



6.18. File structure

Data Integrity risk assessment requires a clear understanding of file structure. The way data is structured within the GXP environment will depend on what the data will be used for and the end user may have this dictated to them by the software/computerised system(s) available. There are many types of file structure, the most common being flat files and relational databases.

Different file structures due to their attributes may require different controls and data review methods and may retain meta data in different ways.

6.19. Validation – for intended purpose (GMP; See also Annex 11, 15)

Computerised systems should comply with regulatory requirements and associated guidance. These should be validated for their intended purpose which requires an understanding of the computerised system's function within a process. For this reason, the acceptance of vendor-supplied validation data in isolation of system configuration and users intended use is not acceptable. In isolation from the intended process or end-user IT infrastructure, vendor testing is likely to be limited to functional verification only and may not fulfil the requirements for performance qualification.

Functional verification demonstrates that the required information is consistently and completely presented. Validation for intended purpose ensures that the steps for generating the custom report accurately reflect those described in the data checking SOP and that the report output is consistent with the procedural steps for performing the subsequent review.

6.20. IT Suppliers and Service Providers (including Cloud providers and virtual service/platforms (also referred to as software as a service SaaS/platform as a service (PaaS) / infrastructure as a service (IaaS)).

Where 'cloud' or 'virtual' services are used, attention should be paid to understanding the service provided, ownership, retrieval, retention and security of data.

The physical location where the data is held, including the impact of any laws applicable to that geographic location, should be considered.

The responsibilities of the contract giver and acceptor should be defined in a technical agreement or contract. This should ensure timely access to data (including metadata and audit trails) to the data owner and national competent authorities upon request. Contracts with providers should define responsibilities for archiving and continued readability of the data throughout the retention period (see archive).

Appropriate arrangements must exist for the restoration of the software/system as per its original validated state, including validation and change control information to permit this restoration.

Business continuity arrangements should be included in the contract, and tested. The need for an audit of the service provider should be based upon risk.



7. Glossary

Acronym or word or phrase	Definition
eCRF	Electronic Case Report Form
ECG	Electrocardiogram
GXP	Good 'X' Practice where 'X' is used as a collective term for GDP – Good Distribution Practice, GCP – Good Clinical practice, GLP – Good Laboratory Practice GMP – Good Manufacturing Practice GPvP – Good Pharmacovigilance Practice
Data Quality	The assurance that data produced is exactly what was intended to be produced and fit for its intended purpose. This incorporates ALCOA
ALCOA	Acronym referring to Attributable, Legible, Contemporaneous, Original and Accurate.
ALCOA +	Acronym referring to Attributable, Legible, Contemporaneous, Original and Accurate 'plus' Complete, Consistent, Enduring, and Available.
DIRA	Data Integrity Risk Assessment
Terminology	The body of terms used with a particular technical application in a subject of study, profession, etc.
Data cleaning	The process of detecting and correcting (or removing) corrupt or inaccurate records from a record set, table, or database and refers to identifying incomplete, incorrect, inaccurate or irrelevant parts of the data and then replacing, modifying, or deleting the dirty or coarse data.
Format	The something is arranged or set out
Directly accessible	At once; without delay
Procedures	Written instructions or other documentation describing process i.e. standard operating procedures (SOP)
Advanced electronic signatures	an electronic signature based upon cryptographic methods of originator authentication, computed by using a set of rules and a set of parameters such that the identity of the signer and the integrity of the data can be verified.
Validated scanning process	A process whereby documents / items are scanned as a process with added controls such as location identifiers and OCR so that each page duplicated does not have to be further checked by a human.



8. References

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MHRA/HRA *DRAFT* Guidance on the use of electronic consent

(<http://www.hra-decisiontools.org.uk/consent/>)

EU Pharmacovigilance legislation: <http://ec.europa.eu/health/human-use/pharmacovigilance>

The Human Medicines Regulations 2012 (Statutory Instrument 2012 No. 1916):

<http://www.legislation.gov.uk/ukxi/2012/1916/contents/made>

EU Good Pharmacovigilance Practice Modules:

http://www.ema.europa.eu/ema/index.jsp?curl=pages/regulation/document_listing/document_listing_000345.jsp&mid=WC0b01ac058058f32c

Revision History

Revision	Publication Month	Reason for changes
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